# Wireless Communications
## Using MATLAB

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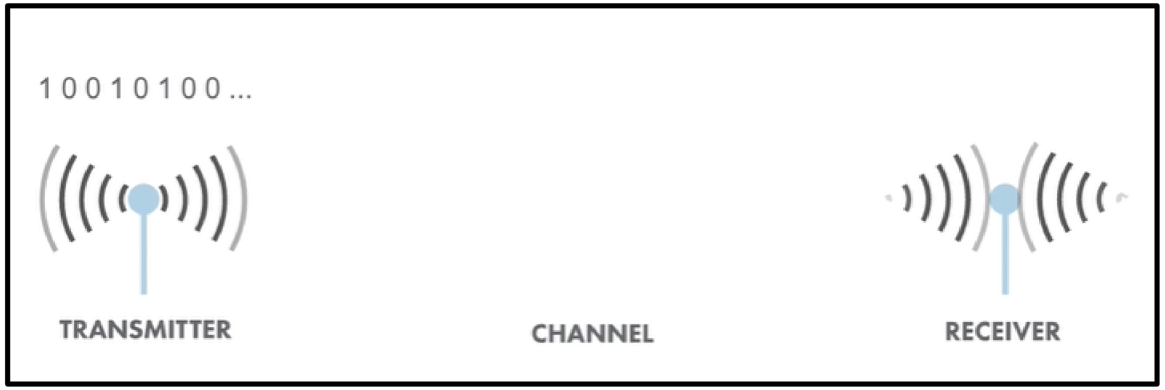
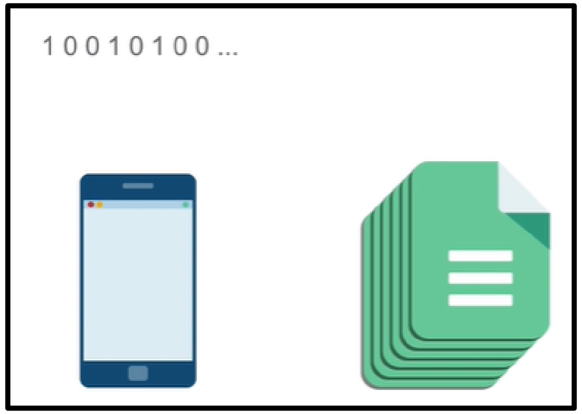
# Most wireless communication systems are digital

- Transmitting binary digits over a channel to a receiver

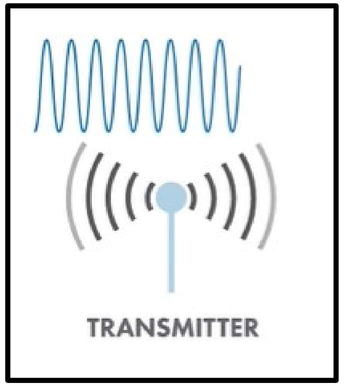



- The binary digits can represent just about anything, a phone call, a document, pictures,..etc.

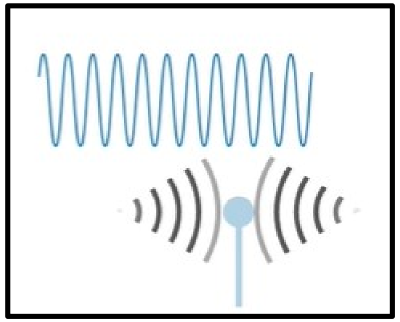


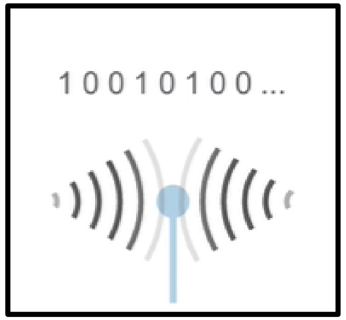
# Most wireless communication systems are digital

- Those bits get transformed into waveforms via modulation




- The receivers then transforms the waveforms back into bits



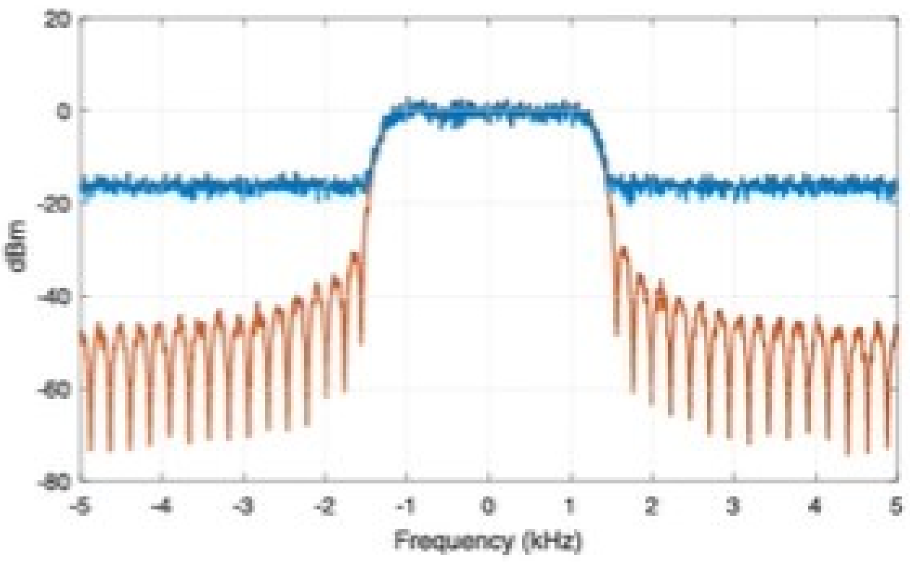


# A typical wireless communication link via QAM

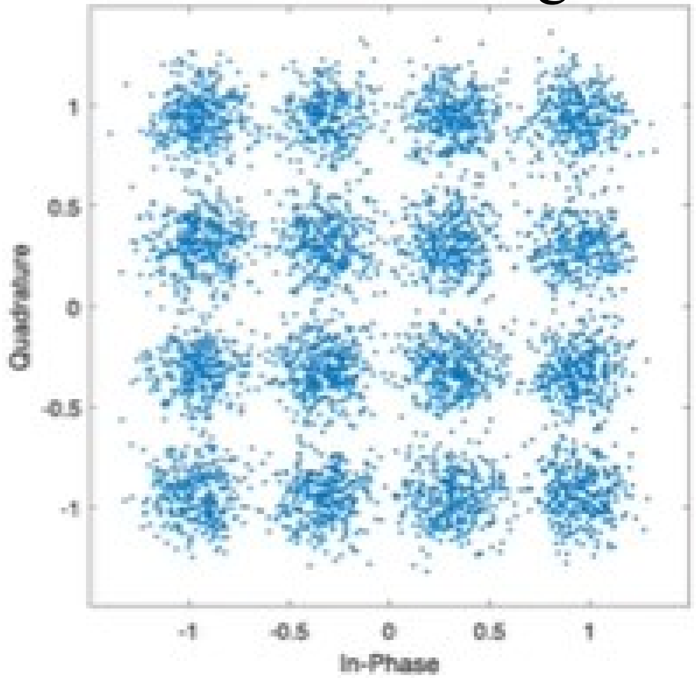
- We will start with quadrature amplitude modulation, QAM
- A single carrier modulation scheme, with a noisy channel, then analyze the link’s performance , Bit Error Rate
- Then we will include filtering to model a more practical QAM

# Next: we will learn to model a multipath channel in wireless applications

- Then, explore how OFDM, for a multicarrier modulation scheme
- OFDM can help compensate for a multipath channel
- Then, we will implement a practical OFDM system



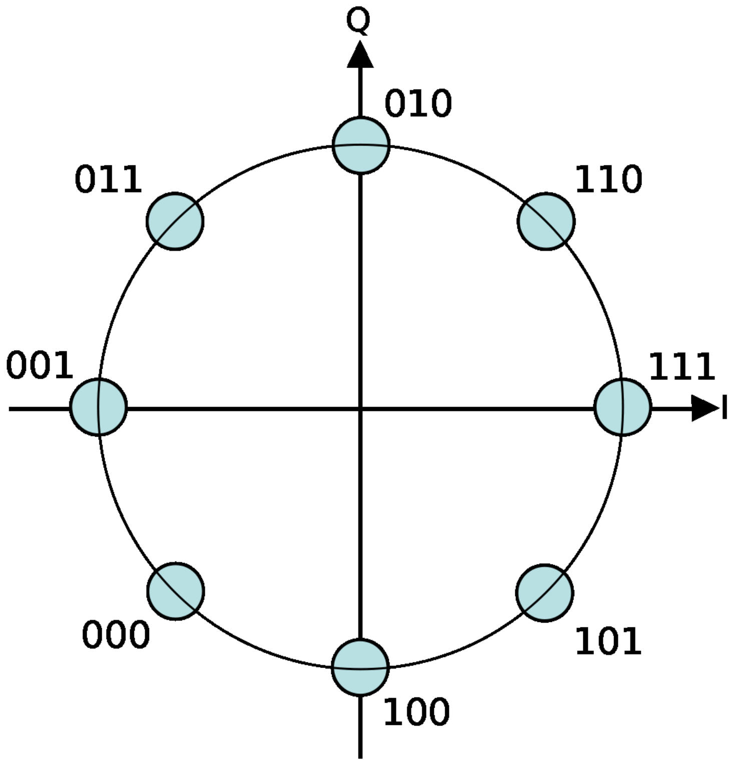

Constellation diagram



# Constellation diagrams

- Constellation diagrams are an important tool in an engineer's to check a digitally modulated radio-frequency (RF) signal modulation

- A constellation diagram is a representation of a signal modulated
  by a digital modulation scheme such as quadrature amplitude modulation or phase-shift keying

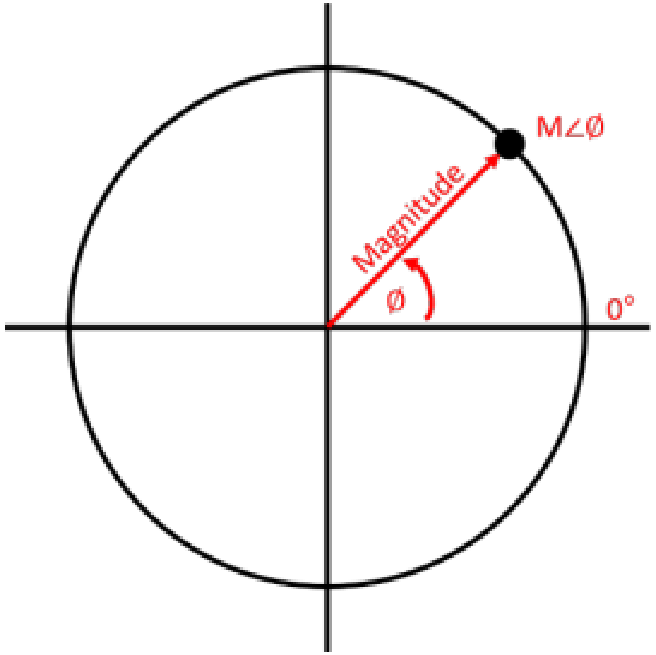



"An 8-PSK. Information transmitted according to the scheme described in the above diagram is encoded as one of 8 "symbols", each representing 3 bits of data. Each symbol is encoded as a different phase shift of the carrier sine wave: 0°, 45°, 90°, 135°, 180°, 225°, 270°, 315°."

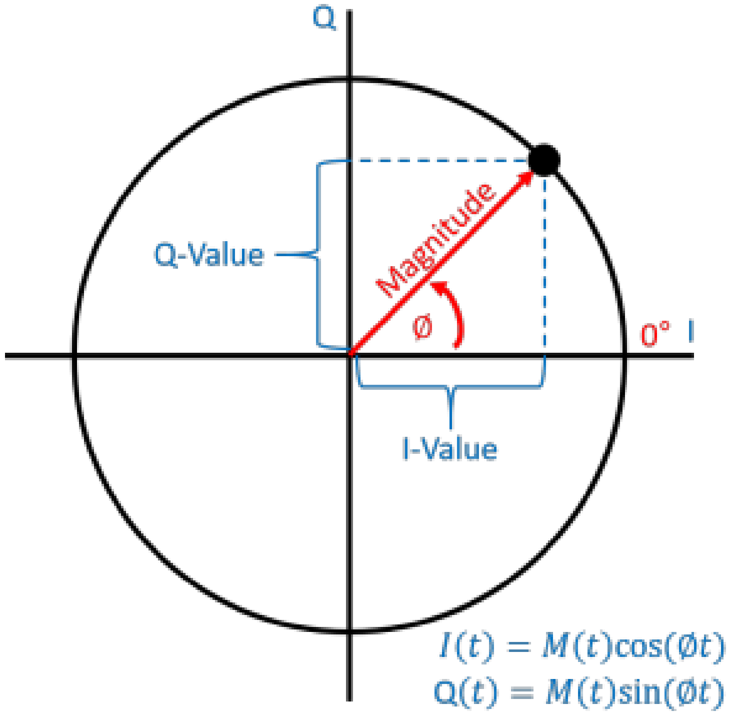
Ref: wiki

# Constellation diagrams

Polar diagram representing amplitude (magnitude) and phase



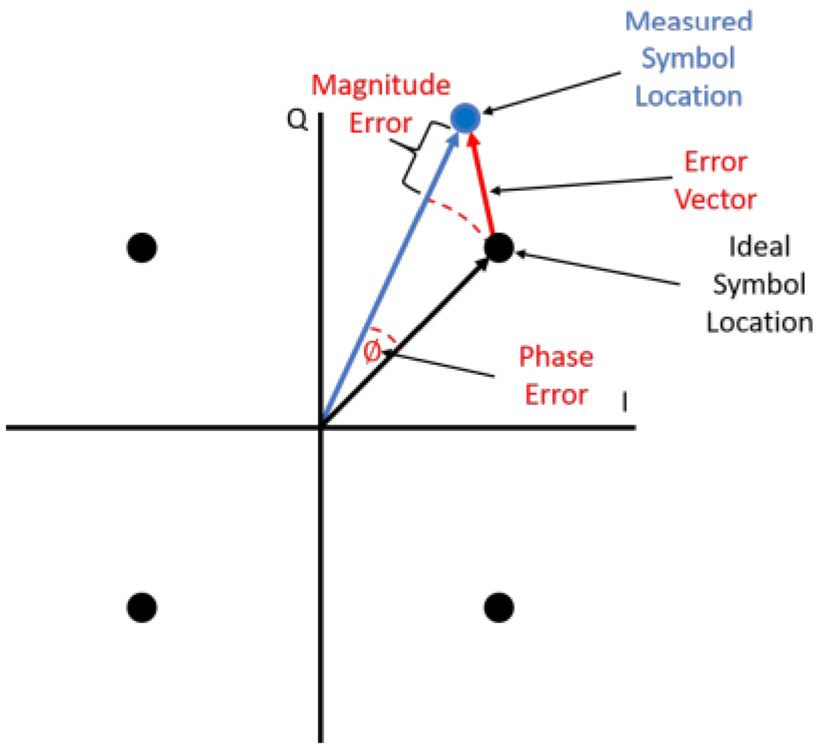

I/Q diagram superimposed on polar diagram




Ref: [nuwaves]

# Constellation diagrams are an invaluable tool for evaluating a digital communication system's performance

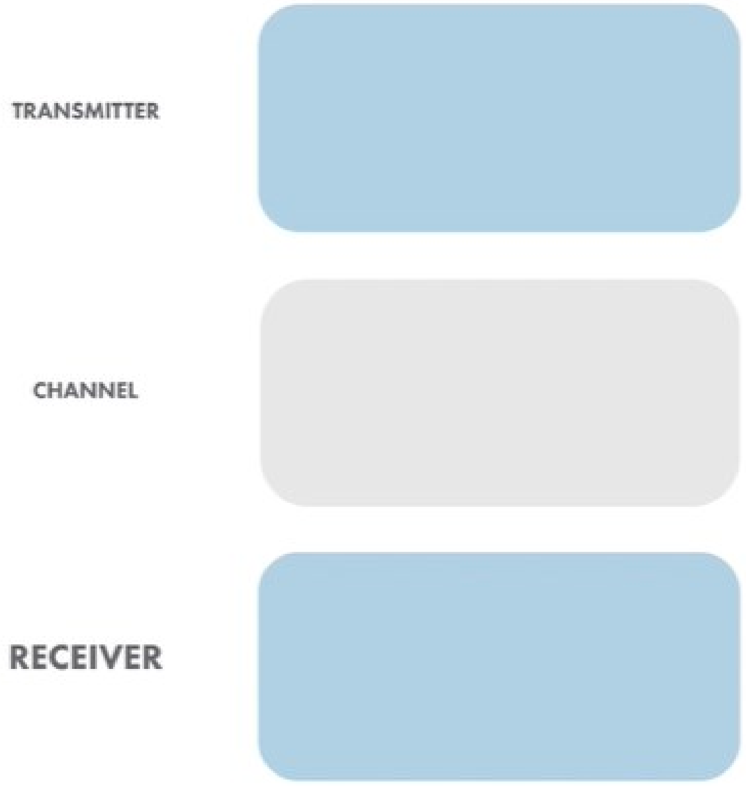
- “These graphical representations of a digitally modulated signal can be used to troubleshoot and isolate various types of signal distortion and interference”
- “Useful for graphically visualizing signal data to quickly identify problems”
- “It is also useful to quantify the disparity between measured and ideal signals as well by calculating the error vector magnitude (EVM)”Ref: nuwaves



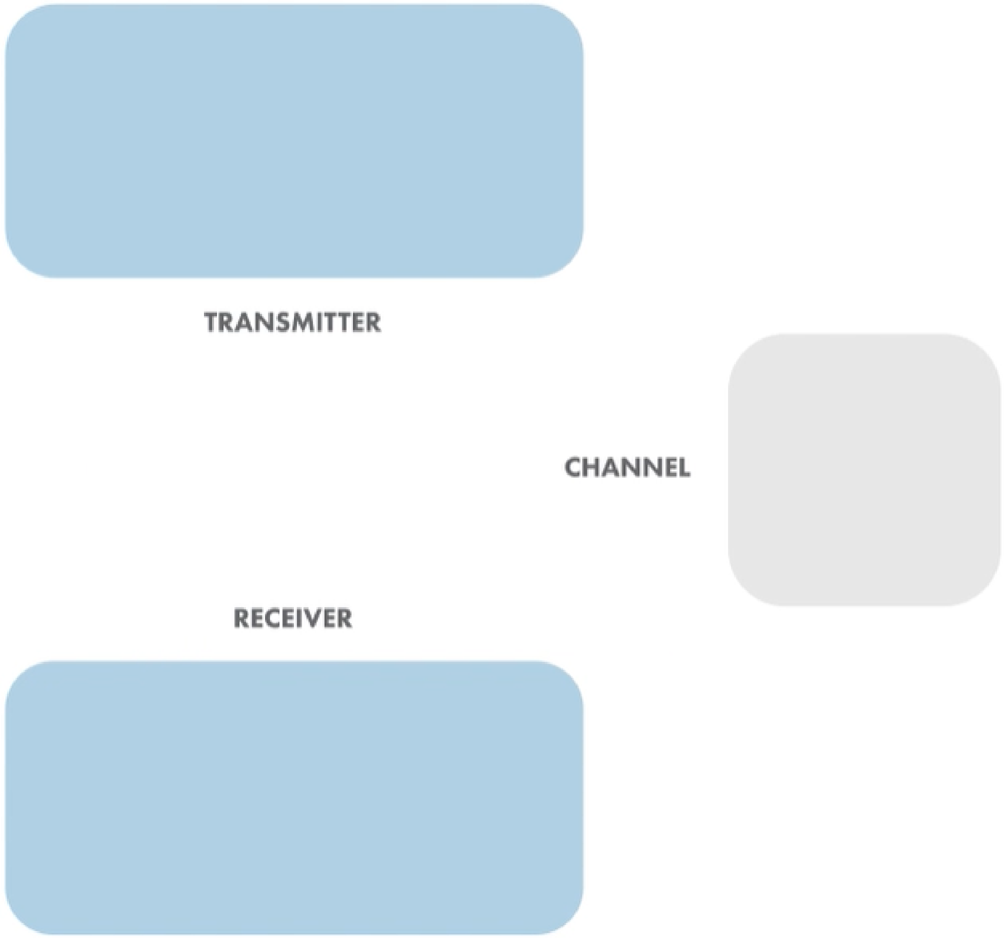

Ref: [nuwaves]

# **Simulate a Basic Digital Communications Link**

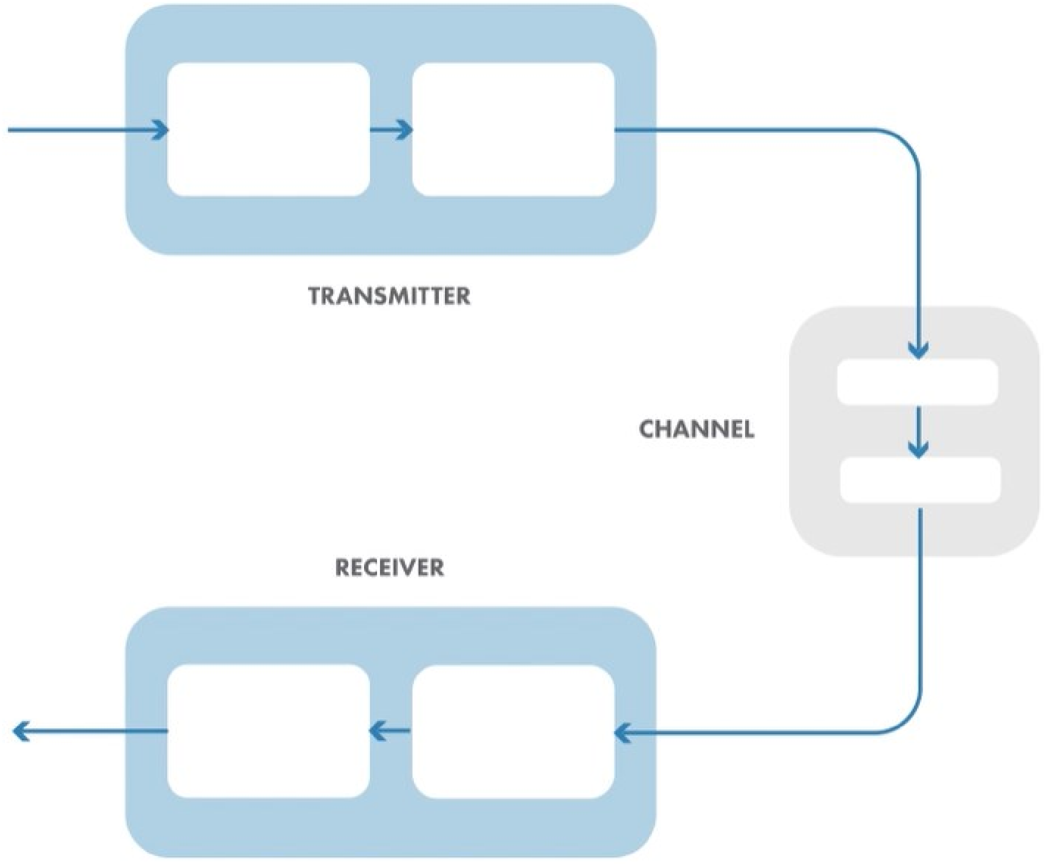
Implement the essential components of a single-carrier digital communications system, including additive white Gaussian noise



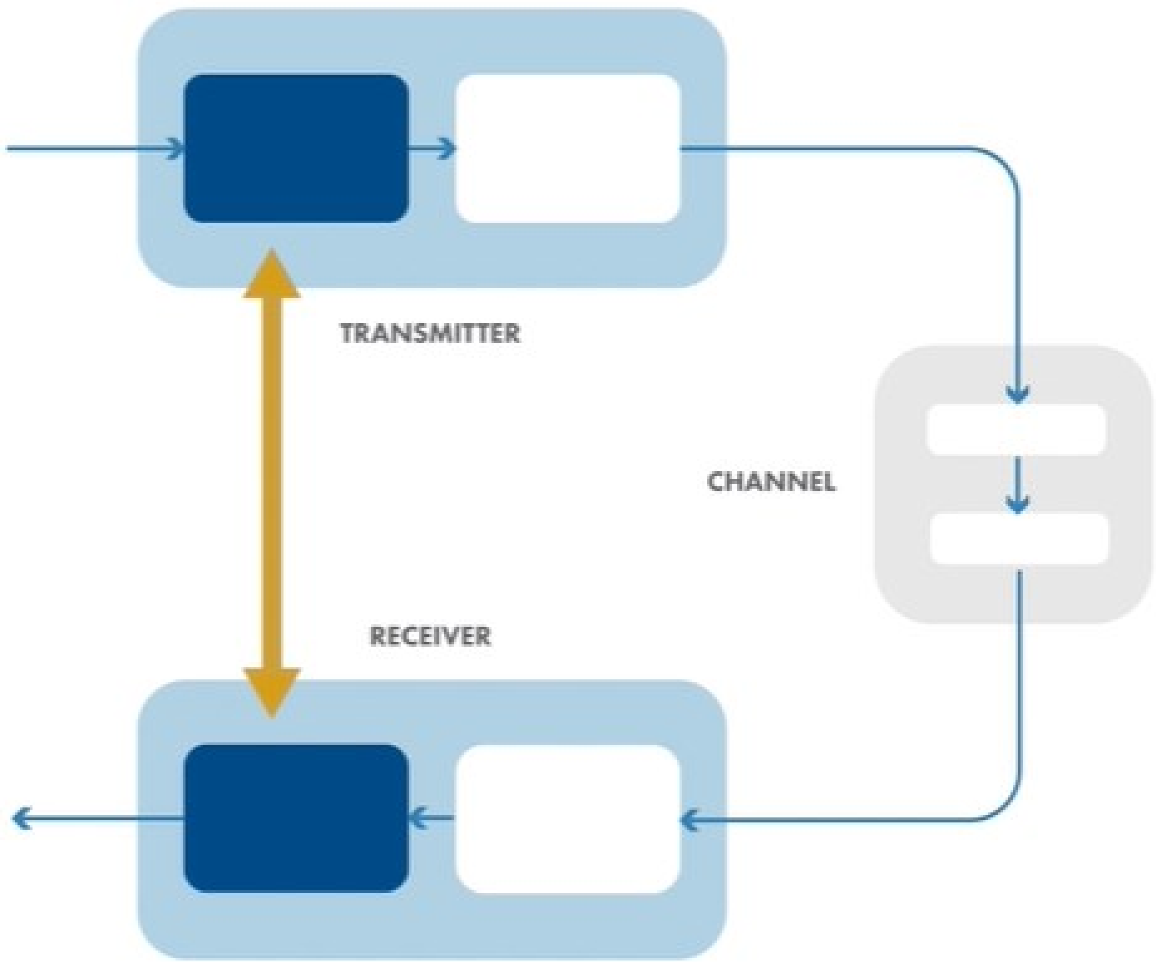
# We build a simulation, you need to represent these components



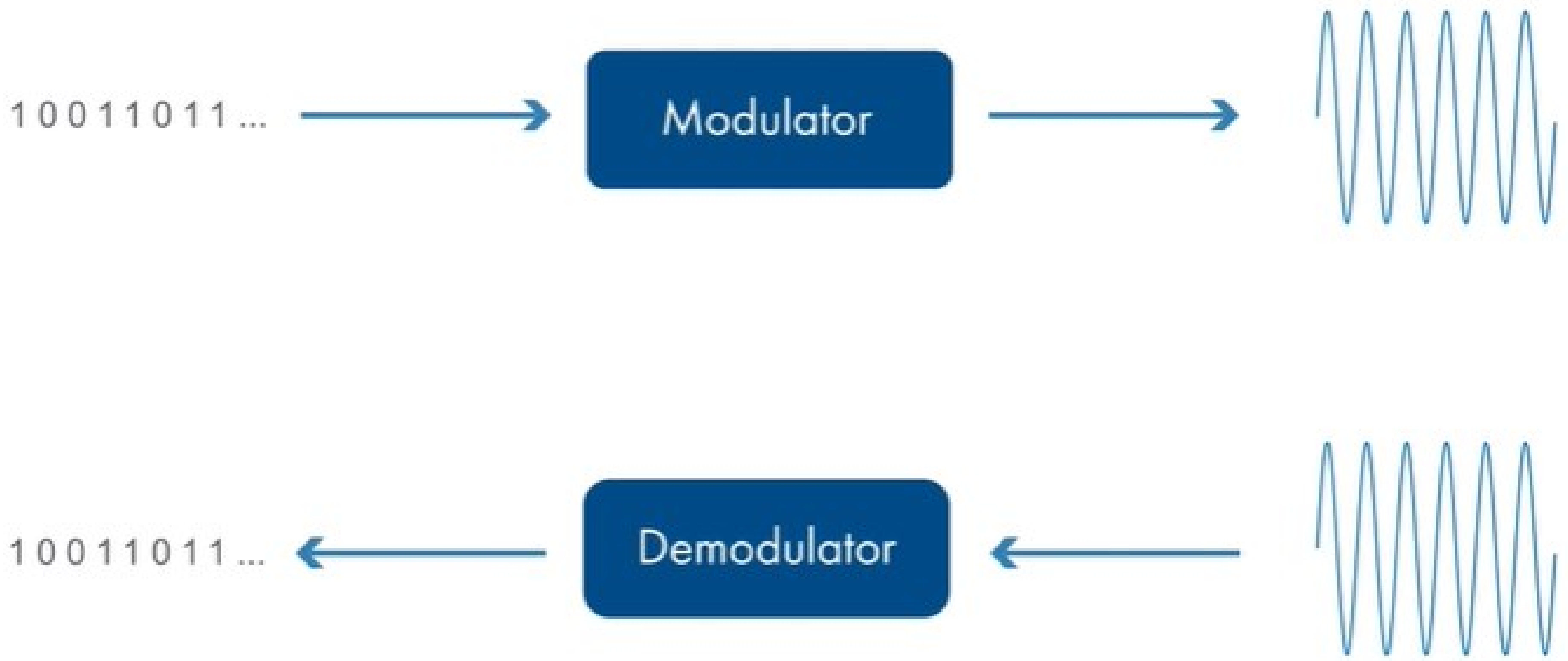
# We build a simulation, you need to represent these components as operations, where each operation's output used as input for the next operation



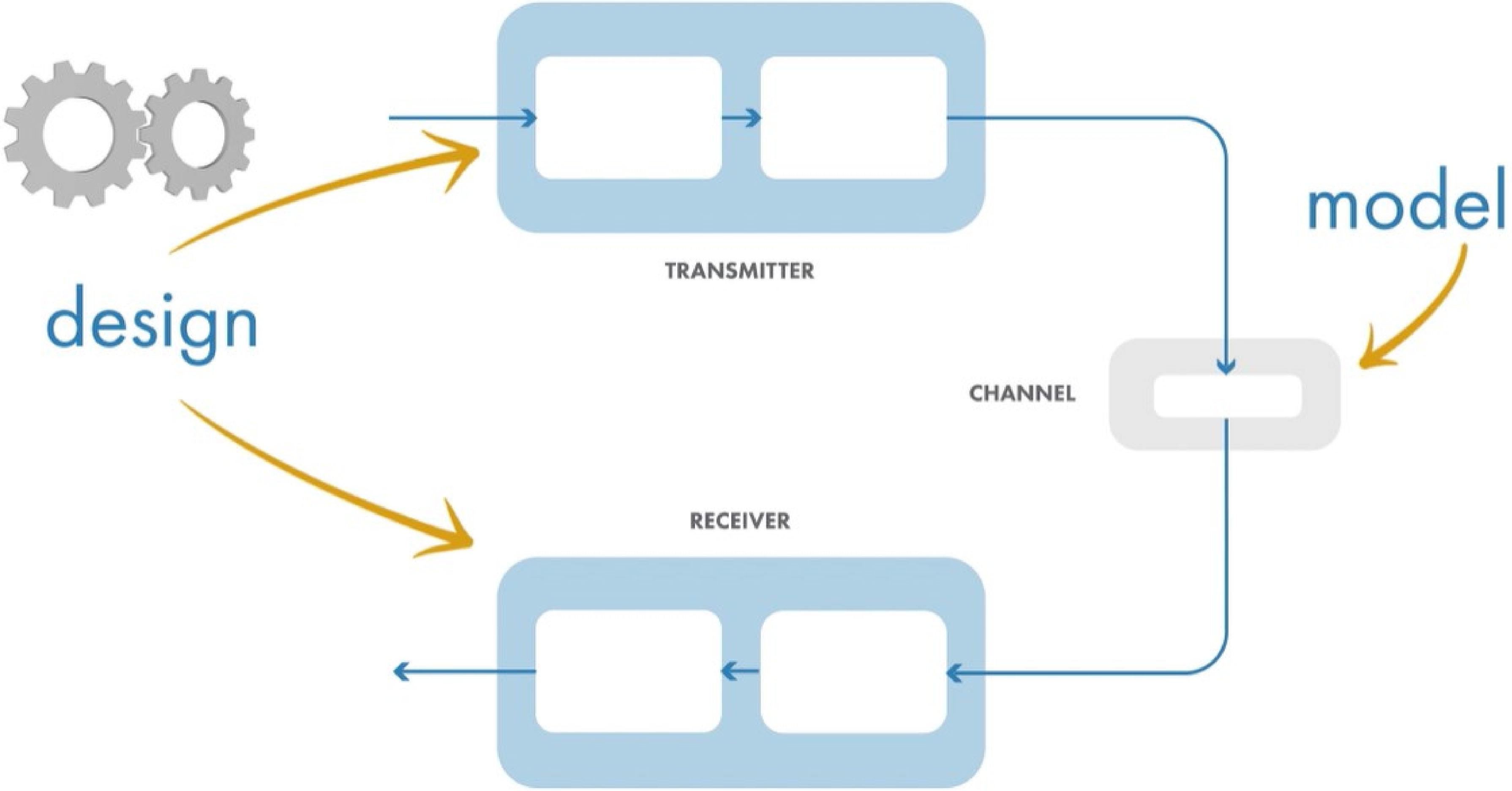
# The transmitter perform an operation and the receiver perform the inverse



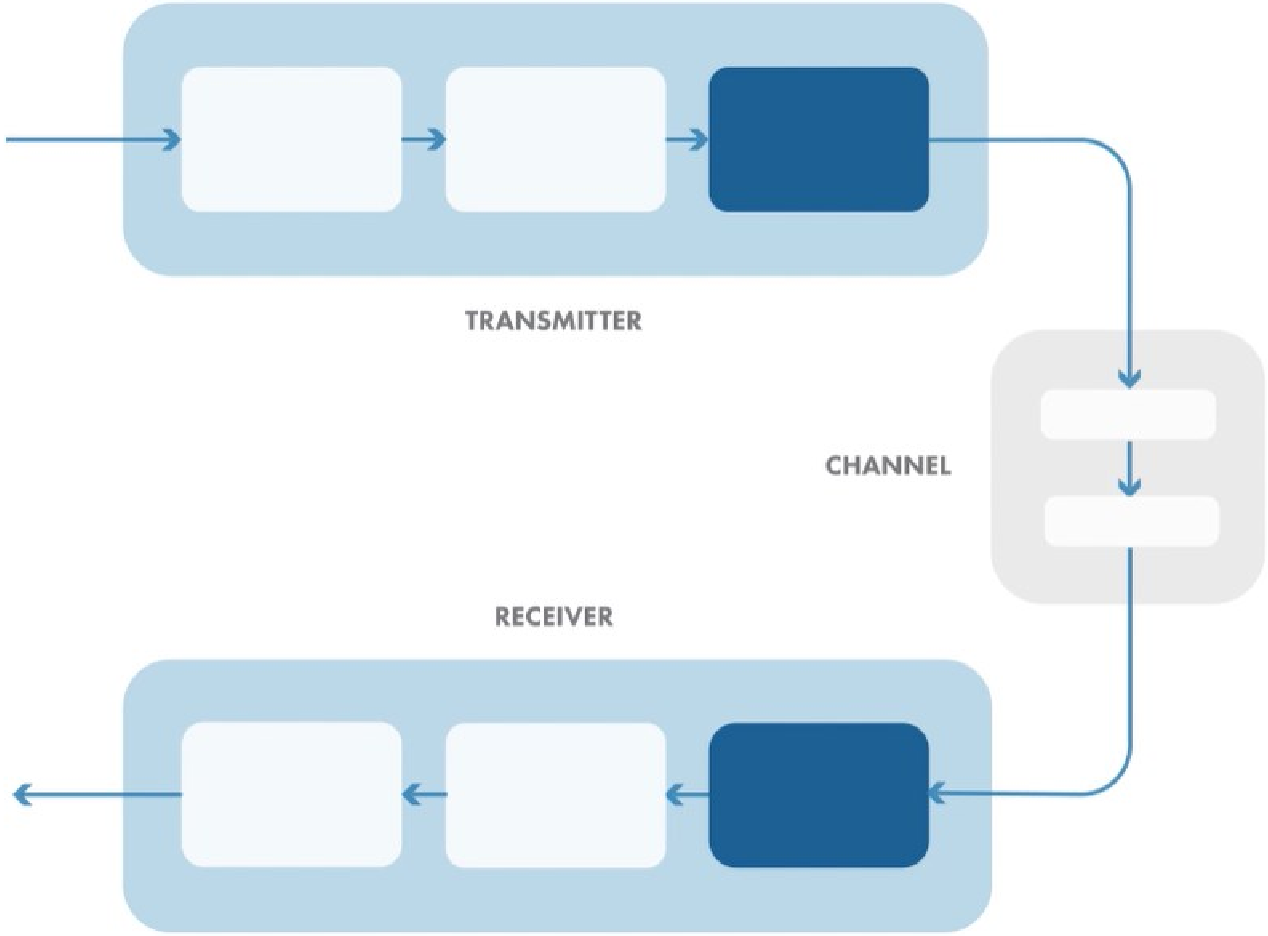
# For example, transmitter converts bits into waveforms via modulation, while receiver convers waveforms into bits



# Both transmitter and receiver are designed to perform a specific operation on an information accurately and efficiently

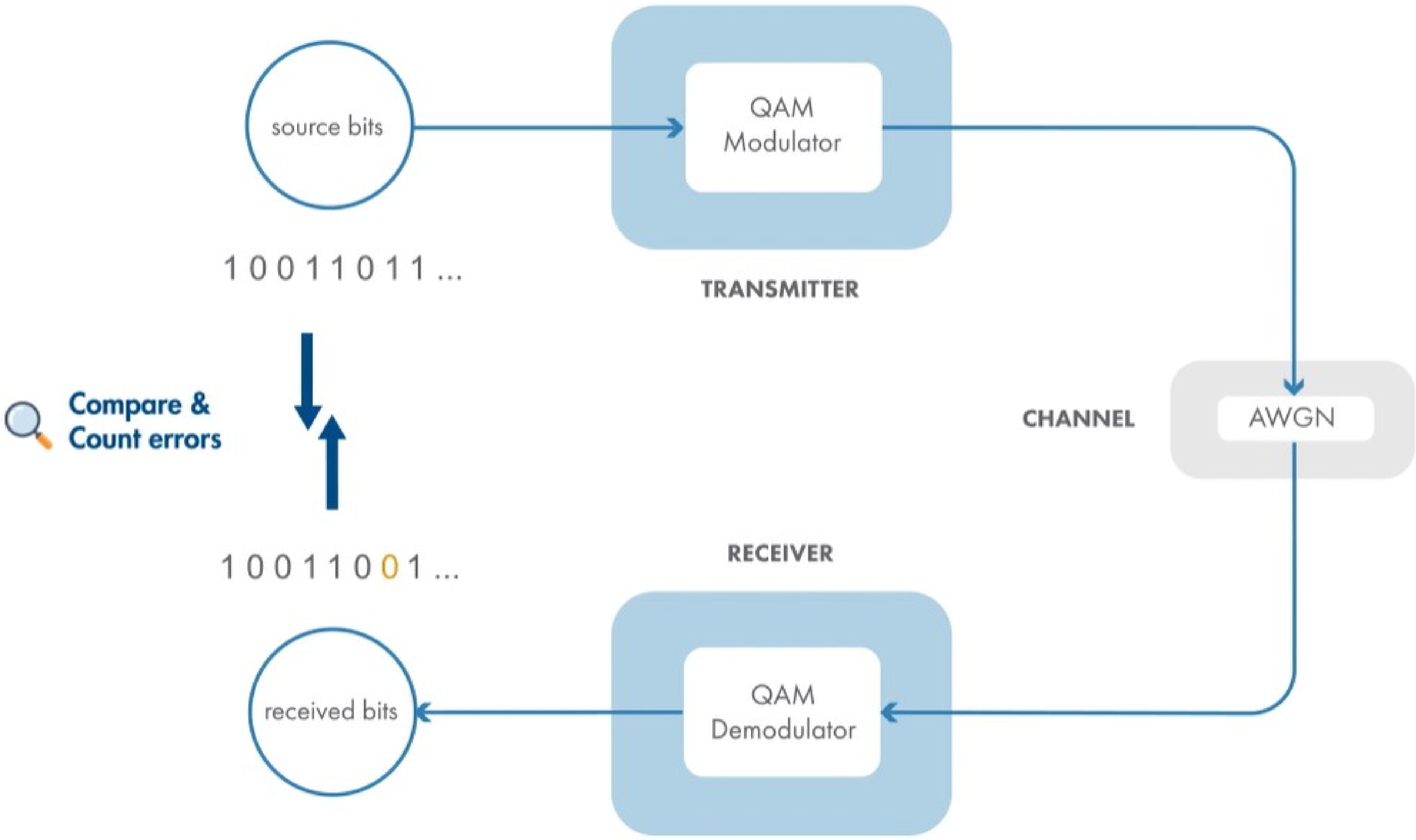


# As you include more impairing effects in your channel model, transmitter and receiver design need to be modified to correct for them



# In the lab, you will model the following communication link

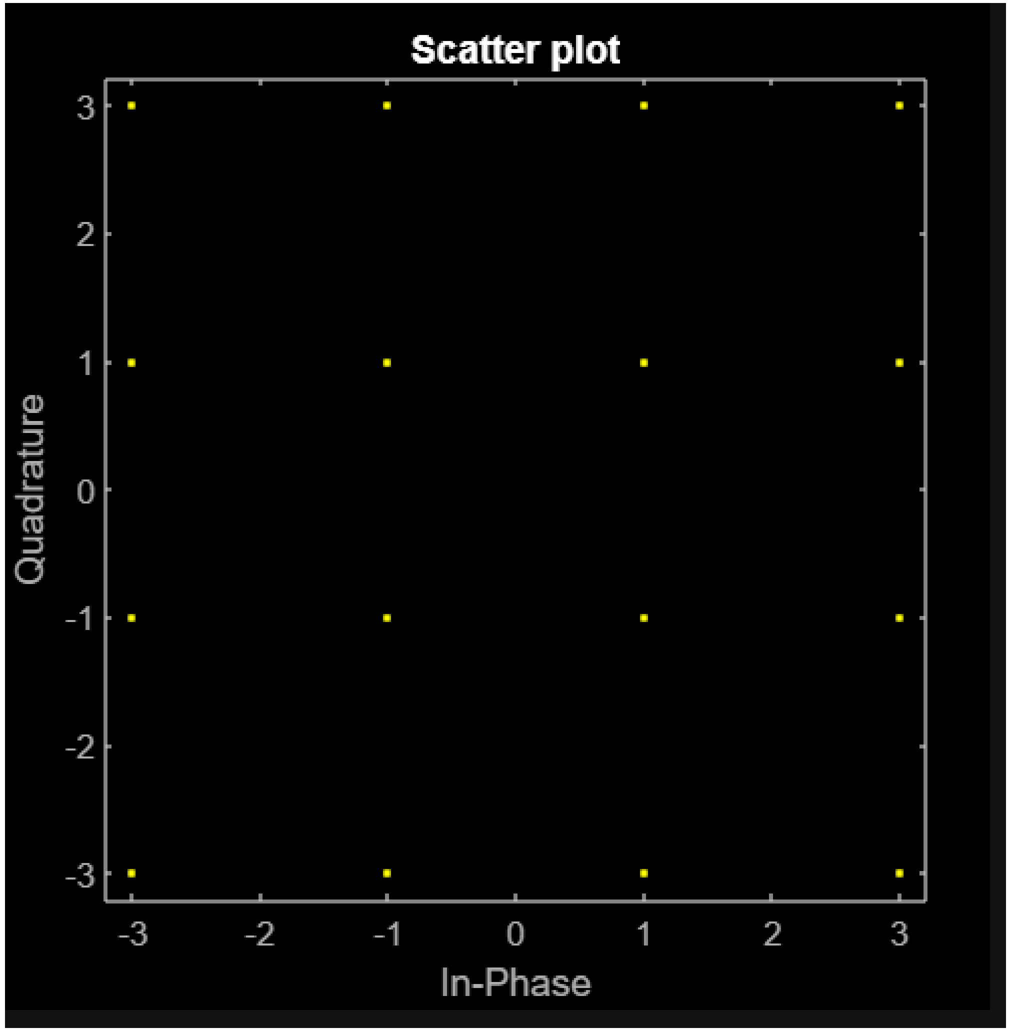
In this activity, you'll simulate a simple back-to-back communications link that uses *quadrature amplitude modulation*, or QAM. You'll generate a source signal of random bits, modulate the source bits using QAM, demodulate the QAM symbols back into bits, and compare the demodulated bits to the source bits to see if they match.



# First step is to create a source of signals

16-QAM is a common single-carrier modulation scheme. It maps 4 input bits to one of 16 complex numbers,
called symbols. For each complex-valued symbol, the real and imaginary parts represent the in-phase and quadrature components, respectively, of a waveform.

The "16" in 16-QAM is
the modulation order. It's useful to store the modulation order in a variable so that it can be used throughout your simulation.

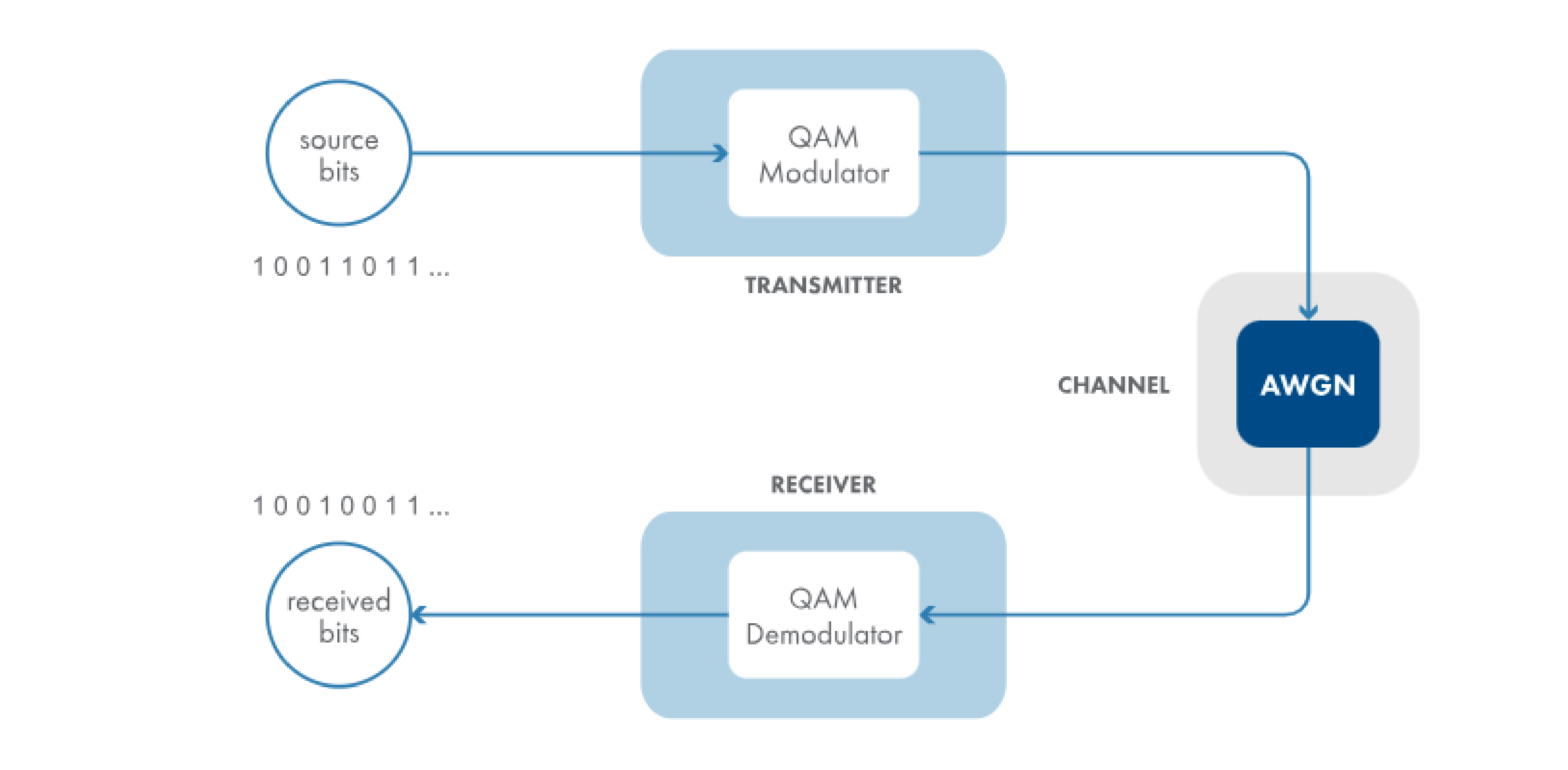


# Then we add noisy channel

A common corruption, or noise source, in the received signal is due to electron motion from the electronics in the receiver. This noise is modeled as part of the channel.

The noise is called *additive white Gaussian noise*, or AWGN, because it's typically *added* to the signal, has a flat (or *white*) spectral density, and has a *Gaussian* probability density function.

In this activity, you'll add noise to your 16-QAM link simulation and note its effects.



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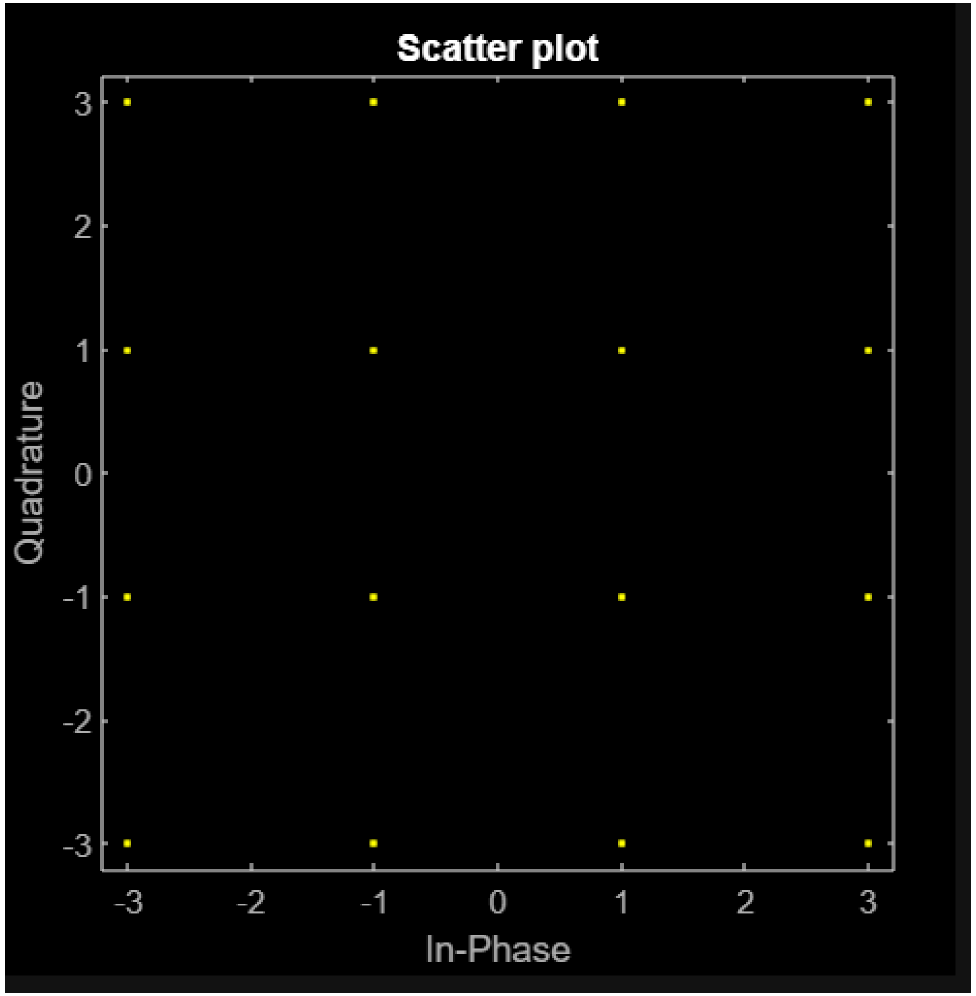
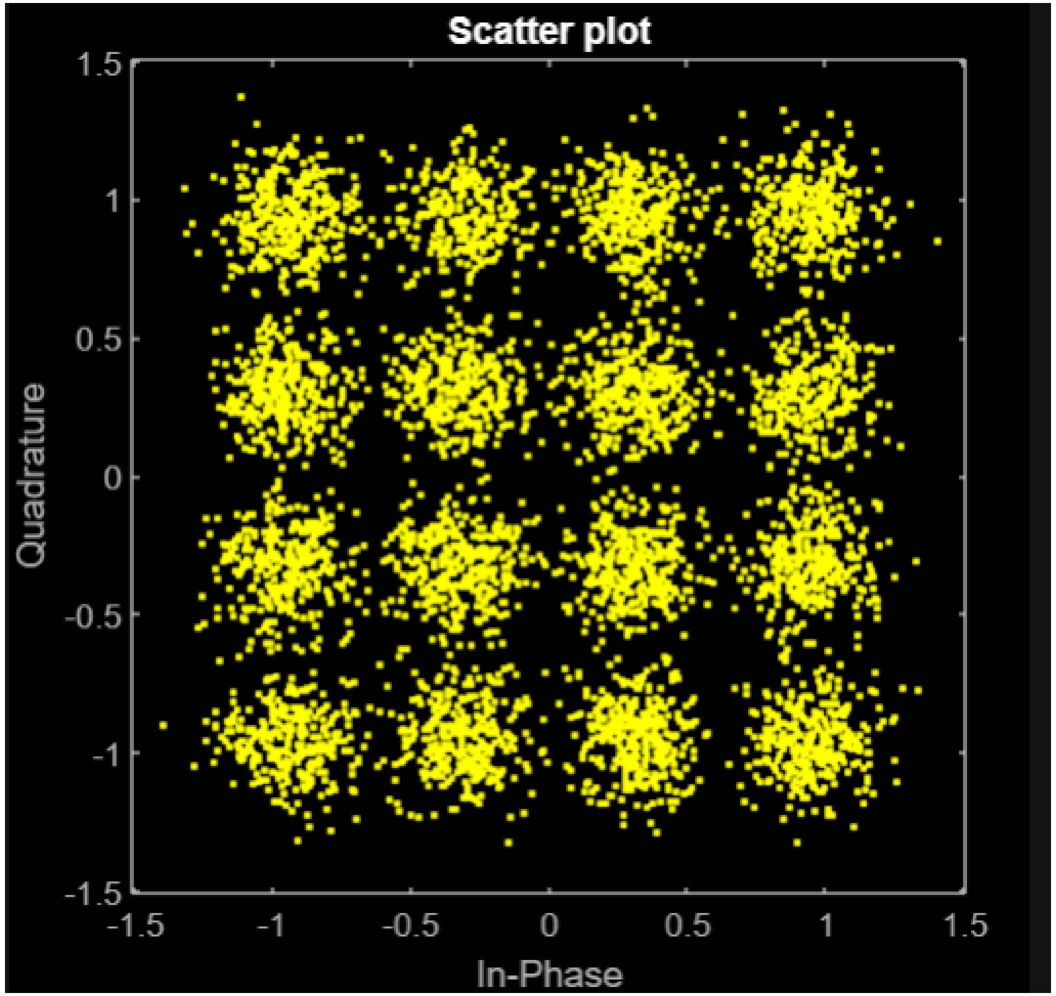
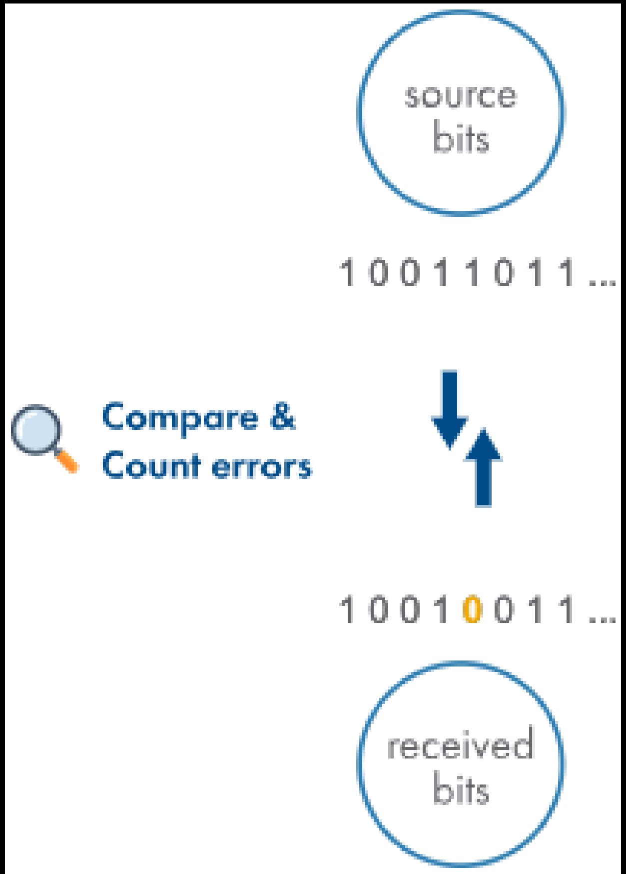


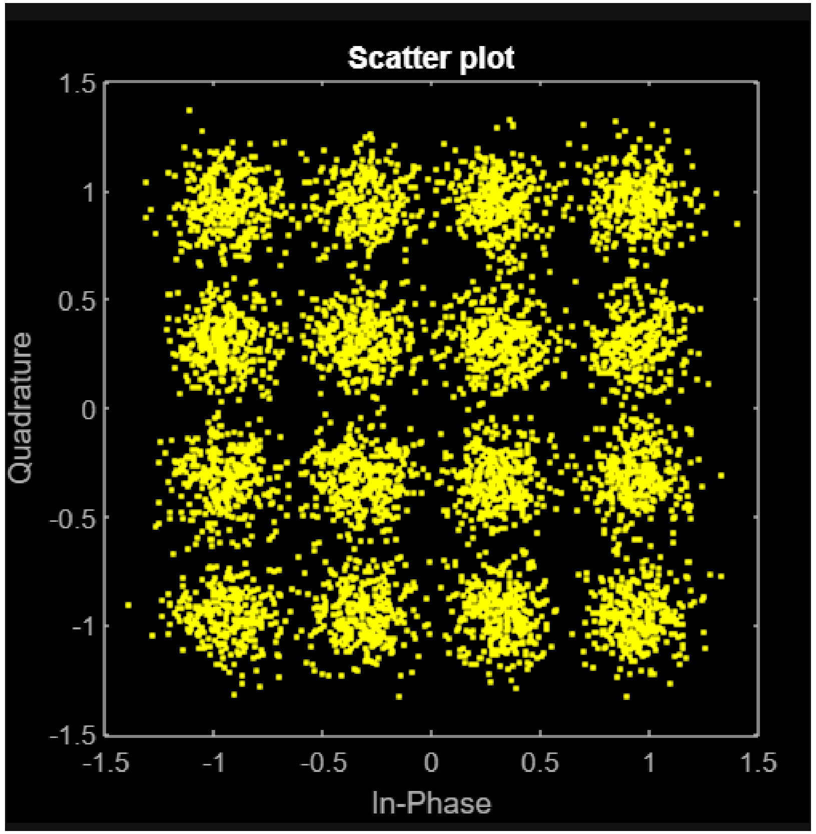
# Calculate the bit error rate

## Calculate the Bit Error Rate

Noise can introduce errors in the received bits. Comparing each bit sent to the corresponding bit received tells you if there are any errors. The fraction of bits that contain bit errors is a key way to measure the performance of a communications system.



# Calculate the bit error rate



# References

[1] https://www.tutorialsmate.com/2021/11/difference-between-data-and-information.html

[2] https://physics.stackexchange.com/questions/314695/electromagnetic-wave-vs-electric-current#:~:text=Electric%20current%20is%20the%20movement,can%20propagate%20through%20a%20vacuum.